

ACHYUT KATIYAR

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EDUCATION

Northeastern University

Master of Science in Computer Science

January 2025 – May 2027

Boston, Massachusetts, USA

SRM University AP

Bachelor of Technology in Computer Science and Engineering

October 2020 – May 2024

Andhra Pradesh, India

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, Java, C, C++, SQL, Solidity, Bash
- **Frameworks:** React, Next.js, Node.js, Express, Spring Boot, Tailwind CSS, Three.js, Prisma, Streamlit, Vitest
- **Cloud:** AWS (Lambda, DynamoDB, API Gateway, Cognito, S3, Amplify), GCP (OAuth2, refresh tokens), Terraform
- **DevOps:** Docker, Kubernetes, GitHub Actions, CI/CD, Vercel, Make, Postman, Git, Linux
- **Databases / APIs:** PostgreSQL, MongoDB, Redis, REST APIs, WebSockets, OpenAI API, Anthropic API

WORK EXPERIENCE

Software Engineer Co-op

Maven Realty

June 2026 – August 2026

Cambridge, Massachusetts, USA

- Processed \$1M+ in lease contracts via a multi-party DocuSign workflow that enforces signing order across tenants, guarantors, landlords, and agents while auto-populating lease tabs from application data.
- Integrated timezone-aware scheduling with per-agent Google Meet rooms, OAuth2, AWS S3 storage, and live property feeds from USPS, WalkScore, and MLS into the Node 22 and Express backend.
- Built a signals-based line-up tool that bundles multiple listings into one shareable link auto-embedded into client messages, then tracked down backend bugs in encrypted Mongoose models, Express 5 query parsing, and calendar-invite MIME formatting.

Teaching Assistant – CS2000

Northeastern University

September 2025 – April 2026

Boston, Massachusetts, USA

- Selected to TA CS2000 (Logic and Computation) across two consecutive semesters for a 500+ student introductory course.
- Held 4 hours of weekly office hours, graded 500+ assignments and exams, and wrote model-solution writeups covering recursion, higher-order functions, and tree abstractions in Python and Pyret.

PROJECTS

MIT Bitcoin Expo 2026 Website | *Next.js, TypeScript, Tailwind CSS, React Three Fiber, shadcn/ui*

- Drew **100k+** visitors to the official 500-attendee conference site with structured JSON-LD SEO and per-speaker landing pages.
- Reduced initial bundle size by lazy-loading a 3D star background via React Three Fiber with SSR-safe hydration boundaries.

PayPhone | *Next.js, TypeScript, AWS, Terraform, Coinbase CDP, x402, Permit2, Anthropic API*

- AI agent bills video calls per-second on Base mainnet via Coinbase CDP x402. Agent signs one Permit2 **upto** \$5 witness, chain settles `floor(active_seconds) × $0.01` per call. **6 on-chain settlements**, 2 on mainnet (\$0.05, \$0.31).
- Hand-rolled the x402 verify-to-settle handshake (vs. CDP's `paymentMiddleware`) so settlement defers to the hangup webhook. Provisioned the AWS stack (Cognito, DynamoDB, scoped IAM) via Terraform as one **apply**.

MIT Optimum Hacknet | *Python, Bash, Streamlit, P2P Networking, Docker* | **Hackathon Winner**

- Proved OptimumP2P holds a flat 161ms P95 latency while GossipSub degrades 140% at 20 msg/sec across a 12-node global cluster.
- Identified shard factor 8 and mesh degree 8–12 as the optimal config via systematic parameter sweep.

SkillBridge AI | *Next.js, TypeScript, PostgreSQL, Prisma, Redis, OpenAI API, ElevenLabs, Docker*

- Released a WCAG 2.1 AA compliant AI voice coaching PWA with GPT-4 and ElevenLabs, hitting a 95+ Lighthouse score.
- Cached frequent GPT-4 completions in Redis to cut OpenAI-to-ElevenLabs voice-chat latency on hot prompts, with Prisma Accelerate for Postgres connection pooling.

Distributed Kanban Board | *C++11, TCP, Vector Clocks, Primary-Backup Replication, Express, Socket.io, React*

- Engineered a real-time collaborative Kanban with C++ primary-backup replication. Hot-standby promotes on a 5-second heartbeat, and the rejoining master receives a full state transfer from it.
- Vector clocks for last-write-wins conflict resolution, a custom TCP binary protocol, and Express + Socket.io as the gateway to the C++ backend. p90 < 200ms / p99 < 500ms under a 50-task concurrent load.

LEADERSHIP / ACHIEVEMENTS

- **Co-Chair, Hackathon & Marketing — MIT Bitcoin Expo 2026:** Judged alongside VCs, angel investors, and senior engineers, and co-led the 36-hour hackathon (47 teams, 150 participants, \$20K prize pool), designing the challenge tracks. Led marketing for the longest-running university-hosted Bitcoin event.
- **Hackathon Prizes (\$3,200+ across 6 wins):** \$1,500 Hacker House Goa 2024 (Nillion + Aptos tracks) | \$600 Move it with Aptos (Okto track) | \$500 Optimum Hacknet @ MIT | \$500 HackSRM 5.0 (Marbelism prize) | \$100 Appathon 2022.